

Weighted Junction-Tree Nodes for Enhanced Interpretability in ADMET Tasks

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Introduction

Motivation

- Accurate molecular representation is crucial for property prediction and drug discovery.
- However, single-graph methods often fail to capture substructural interactions.
- A unified representation is needed to account for unequal atomic importance and substructural complexity.

Goal

- Integrate atomic- and substructural-level knowledge via cross-attention between graphs.
- Achieve interpretable, generalizable, and accurate molecular representations.

Datasets

Pretraining Stage

- ZINC250k[1]:** Large and diverse molecular dataset covering broad chemical space, used to learn generalizable representations.

Downstream Stage

- MoleculeNet[2]:** Classification & Regression Task
- 8:1:1 ratio for train : validation : test
- 5-fold cross-validation
- Random split

Table 1. Details of Benchmark Datasets

Dataset	#Tasks	#Samples	Type	Metric
BBBP	1	2,039	C	ROC-AUC
BACE	1	1,513	C	ROC-AUC
ClinTox	2	1,480	C	ROC-AUC
Tox21	12	7,831	C	ROC-AUC
SIDER	27	1,427	C	ROC-AUC
FreeSolv	1	642	R	RMSE
ESOL	1	1,128	R	RMSE
Lipo	1	4,200	R	RMSE

Methods

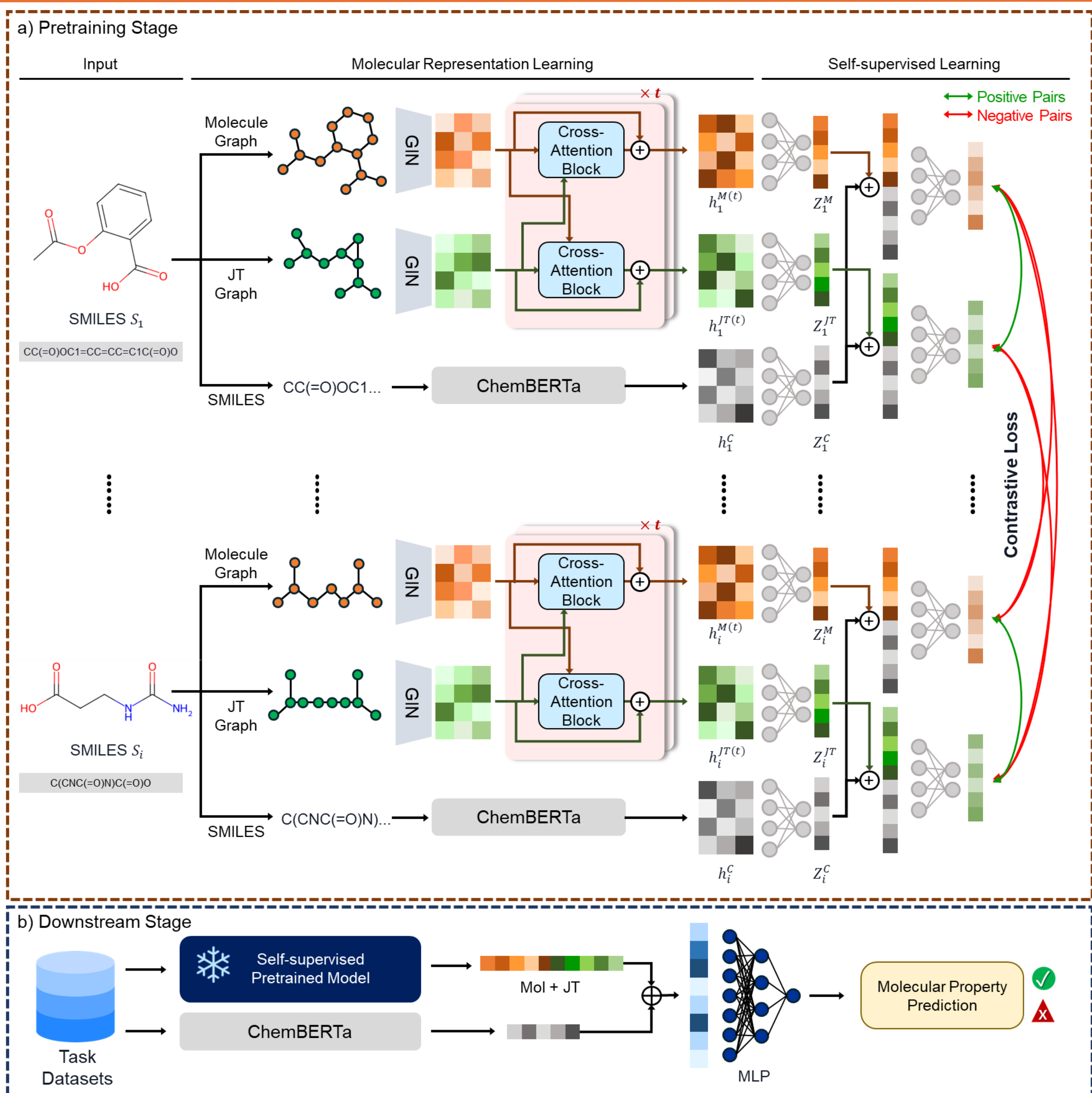


Figure 1. Overview of Our Framework

a) Pretraining Stage

- Input:** $S_i \rightarrow \{G_i^{Mol}, G_i^{JT}, T_i^{SMILES}\}$
- Molecular Representation Learning:** GIN encodes molecules as Mol and JT graph embeddings (atom- and substructure-level). Masked cross-attention exchanges information only across mapped atoms/substructures. ChemBERTa[3] embeddings are fused with Mol and JT embedding to form a unified molecular representation.
- Self-supervised Learning:** Contrastive loss aligns positives (same SMILES) and separates negatives (different SMILES).

b) Downstream Stage

- Input:** Labeled task datasets for molecular property prediction.
- Embedding:** The pretrained model extracts Mol and JT embeddings, and ChemBERTa embeddings.
- Prediction:** The three embeddings are concatenated, and an MLP predicts molecular properties.

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Results

1. Performance Comparison with Baseline

- The proposed model shows strong performance on classification tasks.
- However, its performance on regression tasks remains relatively limited compared to classification.

Table 2. Classification Task (ROC-AUC $\uparrow$)

Dataset	BBBP	BACE	ClinTox	Tox21	SIDER
RNN	0.902 $\pm$ 0.015	—	0.915 $\pm$ 0.009	0.806 $\pm$ 0.007	0.605 $\pm$ 0.009
Transformer	0.944 $\pm$ 0.011	—	0.954 $\pm$ 0.003	0.813 $\pm$ 0.0013	0.602 $\pm$ 0.012
GCN	0.96	—	0.807	0.829	0.638
MPNN	0.910 $\pm$ 0.032	—	0.881 $\pm$ 0.037	0.844 $\pm$ 0.014	0.641 $\pm$ 0.014
GROVER	0.955 $\pm$ 0.003	—	0.929 $\pm$ 0.178	0.842 $\pm$ 0.009	0.663 $\pm$ 0.008
MoCL	0.911	—	0.75	0.824	0.628
MolCLR	0.894 $\pm$ 0.036	0.822 $\pm$ 0.044	0.856 $\pm$ 0.036	0.822 $\pm$ 0.028	0.627 $\pm$ 0.042
GraSeq	0.942 $\pm$ 0.012	—	0.918 $\pm$ 0.005	0.81	0.62
MvMRL	0.962 $\pm$ 0.011	0.891 $\pm$ 0.019	0.975 $\pm$ 0.019	0.845 $\pm$ 0.011	0.653 $\pm$ 0.027
Ours	0.951 $\pm$ 0.017	0.910 $\pm$ 0.018	0.994 $\pm$ 0.002	0.832 $\pm$ 0.011	0.615 $\pm$ 0.007

Table 3. Regression Task (RMSE $\downarrow$)

Dataset	Freesolv	ESOL	Lipo
RNN	1.108 $\pm$ 0.146	0.743 $\pm$ 0.020	0.770 $\pm$ 0.025
Transformer	1.021 $\pm$ 0.102	0.767 $\pm$ 0.079	0.900 $\pm$ 0.0023
GCN	1.4	0.97	—
MPNN	1.242 $\pm$ 0.249	0.702 $\pm$ 0.042	0.645 $\pm$ 0.075
GROVER	1.987 $\pm$ 0.072	0.911 $\pm$ 0.116	0.643 $\pm$ 0.030
MolCL	1.478	0.934	0.742
GraSeq	0.865 $\pm$ 0.032	0.652 $\pm$ 0.039	0.648 $\pm$ 0.041
MvMRL	0.832 $\pm$ 0.128	0.601 $\pm$ 0.055	0.643 $\pm$ 0.018
Ours	1.666 $\pm$ 0.148	0.689 $\pm$ 0.031	0.692 $\pm$ 0.052

Baseline Model

- SMILES-based:** RNN[4], Transformer[5]
- GNN-based:** GCN[6], MPNN[7], GROVER[8], MoCL[9], MolCLR[10]
- Multi-modal:** GraSeq[11], MvMRL[12]

2. Molecular Representation Visualization

- In classification tasks such as BACE and ClinTox, embeddings form well-separated clusters by label, reflecting the model's ability to learn discriminative chemical representations.

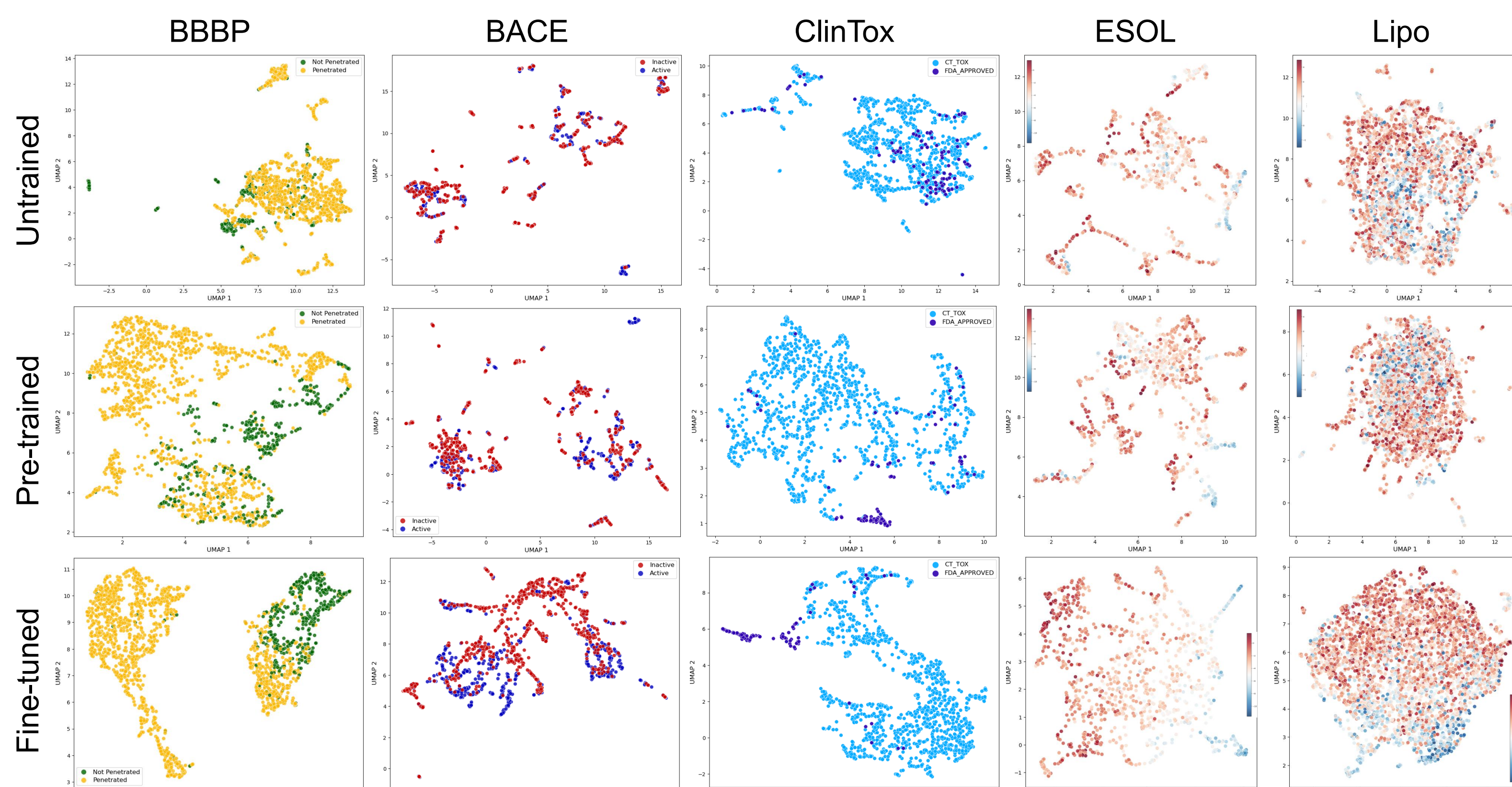


Figure 2. UMAP Visualization of Molecular Representations Across Tasks

3. Learning Substructure-Level Alignments between JT and Molecular Graphs via Cross-attention

- The model learns how each substructure attends differently to its corresponding atoms through cross-attention blocks between JT and Mol graphs.

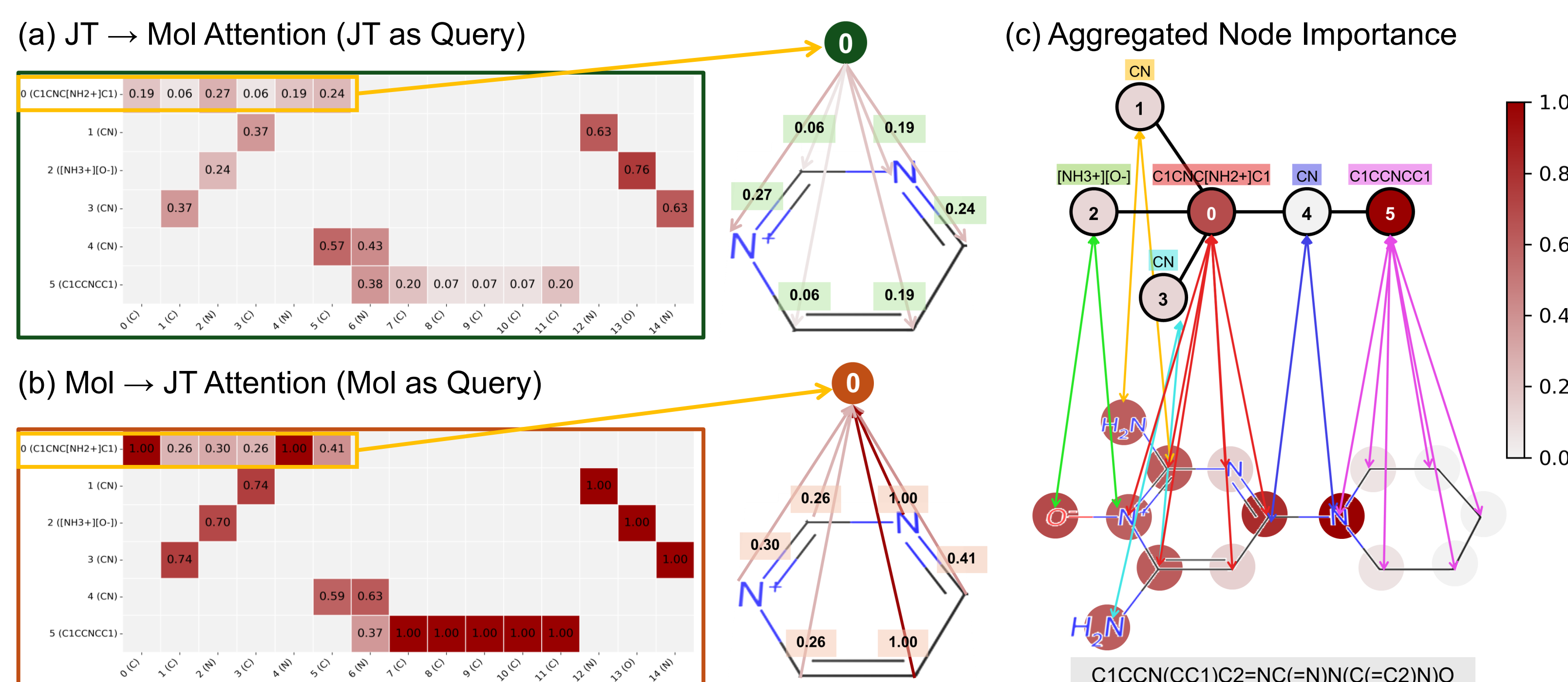


Figure 3. Visualization of Cross-attention between Junction Tree (JT) and Molecular Graph (Mol)

- (a) Cross-attention heatmap from JT to Mol, with visualization of Mol nodes attended by JT node 0. (b) Cross-attention heatmap from Mol to JT, with visualization of Mol nodes connected to JT node 0. Darker lines indicate higher attention weights. (c) Node color represents attention weight (importance), and connecting lines denote the mapping between JT and Mol nodes, visualizing the integrated results of bidirectional attention.

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